Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Dairy Central Heifers

Physical address of dairy:

Number and Street	City	County	Zip Code
10119 W August RD	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 10/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X057-X006-X004-XXXX X057-X006-X005-XXXX X057-X006-X006-XXXX

B. OPERATORS

Borba, Bob

Operator name: Borba, Bob

Telephone no.: Landline: ; Cellular: (209) 321-5444

Mailing Address Number and Street	City	State	Zip Code
6480 Mitchell RD	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Borba, Bob

Legal owner name: Borba, Bob

Telephone no.: Landline: ; Cellular: (209) 321-5444

Mailing Address Number and Street	City	State	Zip Code
6480 Mitchell RD	Hilmar	CA	95324

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	960	680	0	0
Maximum number	0	0	960	680	0	0
Average number	0	0	960	680	0	0
Avg live weight (lbs)	0	0	900	750		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 16,270.02 *tons per reporting period*

Total nitrogen from manure: 155,636.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 108,945.20 *lbs per reporting period*

Total phosphorus from manure: 26,338.40 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 8,095,674 *gallons*

Total nitrogen generated: 126,423.64 *lbs*

Total phosphorus generated: 58,166.07 *lbs*

Total potassium generated: 94,468.67 *lbs*

Total salt generated: 253,735.52 *lbs*

	8,095,674	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	8,095,674	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Dairy Well	Ground water
Irrigation Water - Canal Water	Surface water
Irrigation Water - East Well	Ground water
Irrigation Water - North West Well	Ground water
Irrigation Water - South West Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
#1	72	72	2	process wastewater	X057-X006-X007-XXXX
#2	40	40	2	process wastewater	X057-X006-X003-XXXX
#3	37	37	2	process wastewater	X057-X006-X003-XXXX
#4	37	37	2	process wastewater	X057-X026-X013-XXXX
#5	38	38	2	process wastewater	X057-X026-X013-XXXX
#6	52	52	2	process wastewater	X057-X006-X003-XXXX
Totals for areas that were used for application	276	276	12		
Totals for areas that were not used for application					
Land application area totals	276	276	12		

B. CROPS AND HARVESTS

#1

Field name: #1

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 72 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	1,188.00 *ton*	Dry-weight		65.6	10,400.00	2,700.00	24,500.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.50	118.06	30.65	278.12	987.62

#1

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 72 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	2,025.36 *ton*	Dry-weight		70.4	15,600.00	3,400.00	16,600.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.13	259.79	56.62	276.44	1,082.44

#2

Field name: #2

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	515.20 *ton*	Dry-weight		67.4	11,900.00	2,500.00	21,800.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.88	99.93	20.99	183.07	688.62

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	1,165.60 *ton*	Dry-weight		71.2	16,400.00	3,600.00	16,700.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.14	275.27	60.42	280.30	1,107.79

#3

Field name: #3

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	496.54 *ton*	Dry-weight		68.1	12,400.00	2,700.00	22,700.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.42	106.17	23.12	194.36	693.52

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	1,020.83 *ton*	Dry-weight		69.8	16,500.00	3,500.00	16,500.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.59	274.96	58.33	274.96	1,083.18

#4

Field name: #4

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	463.98 *ton*	Dry-weight		67.2	12,100.00	2,400.00	22,300.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.54	99.54	19.74	183.45	641.65

#4

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	1,099.64 *ton*	Dry-weight		70.7	16,400.00	3,700.00	16,900.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.72	285.62	64.44	294.33	1,166.87

#5

Field name: #5

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	466.26 *ton*	Dry-weight		66.8	13,200.00	2,500.00	22,200.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.27	107.54	20.37	180.87	415.51

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	1,130.12 *ton*	Dry-weight		70.8	17,300.00	3,600.00	16,600.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.74	300.47	62.53	288.31	1,198.40

#6

Field name: #6

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 52 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	941.72 *ton*	Dry-weight		67.4	11,800.00	2,900.00	24,400.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.11	139.33	34.24	288.11	1,015.46

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 52 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	1,490.32 *ton*	Dry-weight		70.5	16,400.00	3,600.00	17,200.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.66	277.31	60.87	290.84	1,132.93

NUTRIENT BUDGET

A. LAND APPLICATIONS

#1 - 11/04/2022: Oats, silage-soft dough

Field name: #1

Crop: Oats, silage-soft dough — Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	0.00	0.00	0.00	49.82	11,617,875.00 *gal*
Application event totals		0.00	0.00	0.00	49.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	0.00	0.00	0.00	50.54	11,786,250.00 *gal*
Application event totals		0.00	0.00	0.00	50.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	143.93	92.98	114.04	385.97	756,842.31 *gal*
Application event totals		143.93	92.98	114.04	385.97	

#1 - 05/25/2023: Corn, silage

Field name: #1

Crop: Corn, silage — Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	294.24	190.09	233.12	789.04	1,547,216.12 *gal*
Application event totals		294.24	190.09	233.12	789.04	

#1 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	0.00	0.00	0.00	51.41	11,988,300.00 *gal*
Application event totals		0.00	0.00	0.00	51.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	0.00	0.00	0.00	51.84	12,089,325.00 *gal*
Application event totals		0.00	0.00	0.00	51.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	0.00	0.00	0.00	52.28	12,190,350.00 *gal*
Application event totals		0.00	0.00	0.00	52.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	0.00	0.00	0.00	52.57	12,257,700.00 *gal*
Application event totals		0.00	0.00	0.00	52.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	0.00	0.00	0.00	52.85	12,325,050.00 *gal*
Application event totals		0.00	0.00	0.00	52.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	0.00	0.00	0.00	52.42	12,224,025.00 *gal*
Application event totals		0.00	0.00	0.00	52.42	

#2 - 11/04/2022: Oats, silage-soft dough

#2 - 11/04/2022: Oats, silage-soft dough

Field name: #2

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.07	0.00	0.00	952.89	3,262,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,708.40	3,262,500.00 *gal*
Application event totals		2.07	0.00	0.00	2,661.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.00	0.00	0.00	920.04	3,150,000.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,649.49	3,150,000.00 *gal*
Application event totals		2.00	0.00	0.00	2,569.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	114.53	73.99	90.74	307.13	334,578.84 *gal*
Application event totals		114.53	73.99	90.74	307.13	

#2 - 05/25/2023: Corn, silage

Field name: #2

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	351.35	71.61	240.72	378.50	725,698.54 *gal*
Application event totals		351.35	71.61	240.72	378.50	

#2 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.21	0.00	0.00	1,018.61	3,487,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,826.23	3,487,500.00 *gal*
Application event totals		2.21	0.00	0.00	2,844.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.24	0.00	0.00	1,029.56	3,525,000.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,845.86	3,525,000.00 *gal*
Application event totals		2.24	0.00	0.00	2,875.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.22	0.00	0.00	1,024.09	3,506,250.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,836.04	3,506,250.00 *gal*
Application event totals		2.22	0.00	0.00	2,860.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.25	0.00	0.00	1,035.04	3,543,750.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,855.68	3,543,750.00 *gal*
Application event totals		2.25	0.00	0.00	2,890.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.24	0.00	0.00	1,029.56	3,525,000.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,845.86	3,525,000.00 *gal*
Application event totals		2.24	0.00	0.00	2,875.43	

#2 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.26	0.00	0.00	1,040.52	3,562,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,865.50	3,562,500.00 *gal*
Application event totals		2.26	0.00	0.00	2,906.02	

#3 - 11/04/2022: Oats, silage-soft dough

Field name: #3

Crop: Oats, silage-soft dough Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.26	0.00	0.00	1,042.00	3,300,000.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,868.15	3,300,000.00 *gal*
Application event totals		2.26	0.00	0.00	2,910.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.29	0.00	0.00	1,053.84	3,337,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,889.38	3,337,500.00 *gal*
Application event totals		2.29	0.00	0.00	2,943.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	112.92	72.95	89.46	302.80	305,128.97 *gal*
Application event totals		112.92	72.95	89.46	302.80	

#3 - 05/25/2023: Corn, silage

#3 - 05/25/2023: Corn, silage

Field name: #3

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	355.16	72.39	243.34	382.60	678,552.33 *gal*
Application event totals		355.16	72.39	243.34	382.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.19	0.00	0.00	1,006.47	3,187,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,804.47	3,187,500.00 *gal*
Application event totals		2.19	0.00	0.00	2,810.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.21	0.00	0.00	1,018.32	3,225,000.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,825.69	3,225,000.00 *gal*
Application event totals		2.21	0.00	0.00	2,844.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.24	0.00	0.00	1,030.16	3,262,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,846.92	3,262,500.00 *gal*
Application event totals		2.24	0.00	0.00	2,877.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.19	0.00	0.00	1,006.47	3,187,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,804.47	3,187,500.00 *gal*
Application event totals		2.19	0.00	0.00	2,810.94	

#3 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.21	0.00	0.00	1,018.32	3,225,000.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,825.69	3,225,000.00 *gal*
Application event totals		2.21	0.00	0.00	2,844.01	

08/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.17	0.00	0.00	1,000.55	3,168,750.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,793.85	3,168,750.00 *gal*
Application event totals		2.17	0.00	0.00	2,794.41	

#4 - 11/05/2022: Oats, silage-soft dough

Field name: #4

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.25	0.00	0.00	1,036.08	3,281,250.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,857.54	3,281,250.00 *gal*
Application event totals		2.25	0.00	0.00	2,893.61	

04/19/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.21	0.00	0.00	1,018.32	3,225,000.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,825.69	3,225,000.00 *gal*
Application event totals		2.21	0.00	0.00	2,844.01	

#4 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	111.22	71.85	88.12	298.26	300,548.49 *gal*
Application event totals		111.22	71.85	88.12	298.26	

#4 - 05/26/2023: Corn, silage

Field name: #4

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	349.89	71.31	239.72	376.92	668,481.66 *gal*
Application event totals		349.89	71.31	239.72	376.92	

05/16/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.19	0.00	0.00	1,006.47	3,187,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,804.47	3,187,500.00 *gal*
Application event totals		2.19	0.00	0.00	2,810.94	

06/27/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.21	0.00	0.00	1,018.32	3,225,000.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,825.69	3,225,000.00 *gal*
Application event totals		2.21	0.00	0.00	2,844.01	

#4 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.24	0.00	0.00	1,030.16	3,262,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,846.92	3,262,500.00 *gal*
Application event totals		2.24	0.00	0.00	2,877.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.20	0.00	0.00	1,012.40	3,206,250.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,815.08	3,206,250.00 *gal*
Application event totals		2.20	0.00	0.00	2,827.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.21	0.00	0.00	1,018.32	3,225,000.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,825.69	3,225,000.00 *gal*
Application event totals		2.21	0.00	0.00	2,844.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.24	0.00	0.00	1,030.16	3,262,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,846.92	3,262,500.00 *gal*
Application event totals		2.24	0.00	0.00	2,877.08	

#5 - 11/05/2022: Oats, silage-soft dough

Field name: #5

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

#5 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	1.98	0.00	0.00	910.81	2,962,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,632.96	2,962,500.00 *gal*
Application event totals		1.98	0.00	0.00	2,543.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.03	0.00	0.00	933.87	3,037,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,674.30	3,037,500.00 *gal*
Application event totals		2.03	0.00	0.00	2,608.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	107.06	69.16	84.82	287.09	297,114.75 *gal*
Application event totals		107.06	69.16	84.82	287.09	

#5 - 05/26/2023: Corn, silage

Field name: #5

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	342.55	69.82	234.70	369.02	672,148.30 *gal*
Application event totals		342.55	69.82	234.70	369.02	

#5 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.23	0.00	0.00	1,026.11	3,337,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,839.66	3,337,500.00 *gal*
Application event totals		2.23	0.00	0.00	2,865.77	

06/28/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.18	0.00	0.00	1,003.05	3,262,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,798.32	3,262,500.00 *gal*
Application event totals		2.18	0.00	0.00	2,801.37	

07/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.22	0.00	0.00	1,020.34	3,318,750.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,829.33	3,318,750.00 *gal*
Application event totals		2.22	0.00	0.00	2,849.67	

07/22/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.24	0.00	0.00	1,031.87	3,356,250.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,850.00	3,356,250.00 *gal*
Application event totals		2.24	0.00	0.00	2,881.87	

08/04/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.25	0.00	0.00	1,037.63	3,375,000.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,860.33	3,375,000.00 *gal*
Application event totals		2.25	0.00	0.00	2,897.97	

#5 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.22	0.00	0.00	1,020.34	3,318,750.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,829.33	3,318,750.00 *gal*
Application event totals		2.22	0.00	0.00	2,849.67	

#6 - 11/05/2022: Oats, silage-soft dough

Field name: #6

Crop: Oats, silage-soft dough Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.05	0.00	0.00	943.63	4,200,000.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,691.79	4,200,000.00 *gal*
Application event totals		2.05	0.00	0.00	2,635.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.08	0.00	0.00	956.26	4,256,250.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,714.45	4,256,250.00 *gal*
Application event totals		2.08	0.00	0.00	2,670.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	171.52	110.81	135.89	459.95	651,380.28 *gal*
Application event totals		171.52	110.81	135.89	459.95	

#6 - 05/26/2023: Corn, silage

#6 - 05/26/2023: Corn, silage

Field name: #6

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	304.92	196.98	241.58	817.67	1,157,983.88 *gal*
Application event totals		304.92	196.98	241.58	817.67	

05/16/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.16	0.00	0.00	994.18	4,425,000.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,782.42	4,425,000.00 *gal*
Application event totals		2.16	0.00	0.00	2,776.60	

06/27/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.18	0.00	0.00	1,002.60	4,462,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,797.53	4,462,500.00 *gal*
Application event totals		2.18	0.00	0.00	2,800.13	

07/09/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.21	0.00	0.00	1,019.45	4,537,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,827.74	4,537,500.00 *gal*
Application event totals		2.21	0.00	0.00	2,847.19	

07/21/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.20	0.00	0.00	1,015.24	4,518,750.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,820.18	4,518,750.00 *gal*
Application event totals		2.20	0.00	0.00	2,835.42	

#6 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.22	0.00	0.00	1,023.67	4,556,250.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,835.29	4,556,250.00 *gal*
Application event totals		2.22	0.00	0.00	2,858.95	

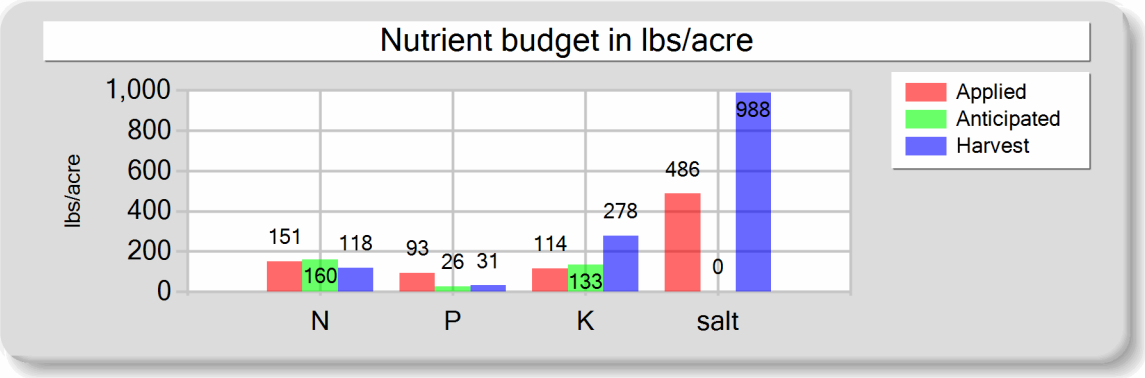
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - East Well	Ground water	2.18	0.00	0.00	1,002.60	4,462,500.00 *gal*
Irrigation Water - North West Well	Ground water	0.00	0.00	0.00	1,797.53	4,462,500.00 *gal*
Application event totals		2.18	0.00	0.00	2,800.13	

B. NUTRIENT BUDGET

#1 - 11/04/2022: Oats, silage-soft dough

Field name: #1 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	143.93	92.98	114.04	385.97
Fresh water	0.00	0.00	0.00	100.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	150.93	92.98	114.04	486.33
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	118.06	30.65	278.12	987.62
Nutrient balance	32.87	62.33	-164.09	-501.29
Applied to removed ratio	1.28	3.03	0.41	0.49

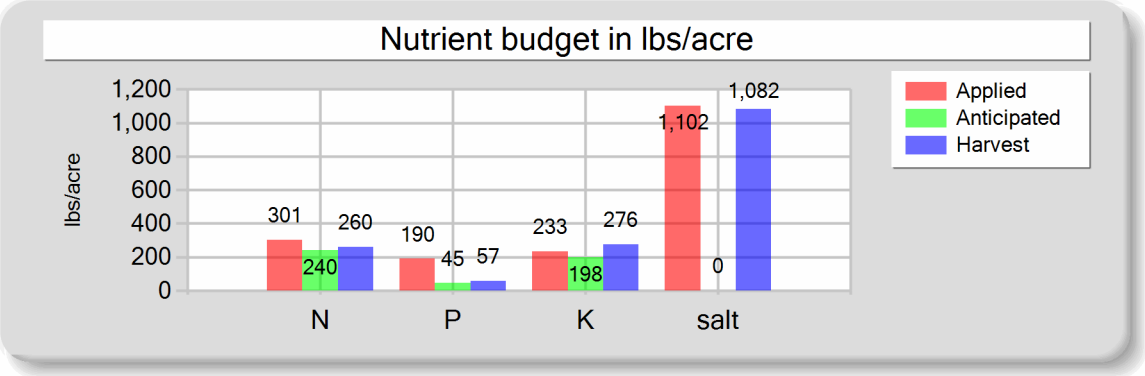
Fresh water applied	
23,404,125.00	*gallons*
861.89	*acre-inches*
11.97	*inches/acre*

Process wastewater applied	
756,842.31	*gallons*
27.87	*acre-inches*
0.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

#1 - 05/25/2023: Corn, silage

Field name: #1 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	294.24	190.09	233.12	789.04
Fresh water	0.00	0.00	0.00	313.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	301.24	190.09	233.12	1,102.41
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	259.79	56.62	276.44	1,082.44
Nutrient balance	41.45	133.47	-43.31	19.97
Applied to removed ratio	1.16	3.36	0.84	1.02

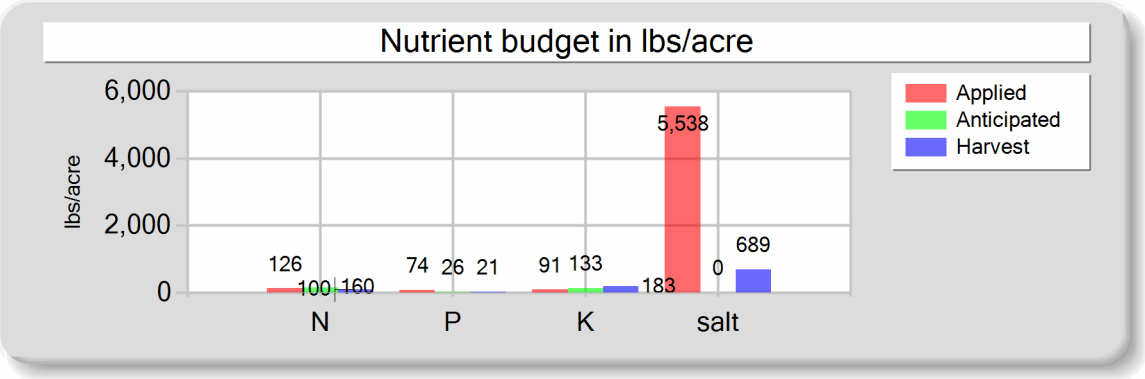
Fresh water applied	
73,074,750.00	*gallons*
2,691.09	*acre-inches*
37.38	*inches/acre*

Process wastewater applied	
1,547,216.12	*gallons*
56.98	*acre-inches*
0.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

#2 - 11/04/2022: Oats, silage-soft dough

Field name: #2 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	114.53	73.99	90.74	307.13
Fresh water	4.07	0.00	0.00	5,230.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	125.60	73.99	90.74	5,537.96
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	99.93	20.99	183.07	688.62
Nutrient balance	25.66	53.00	-92.33	4,849.34
Applied to removed ratio	1.26	3.52	0.50	8.04

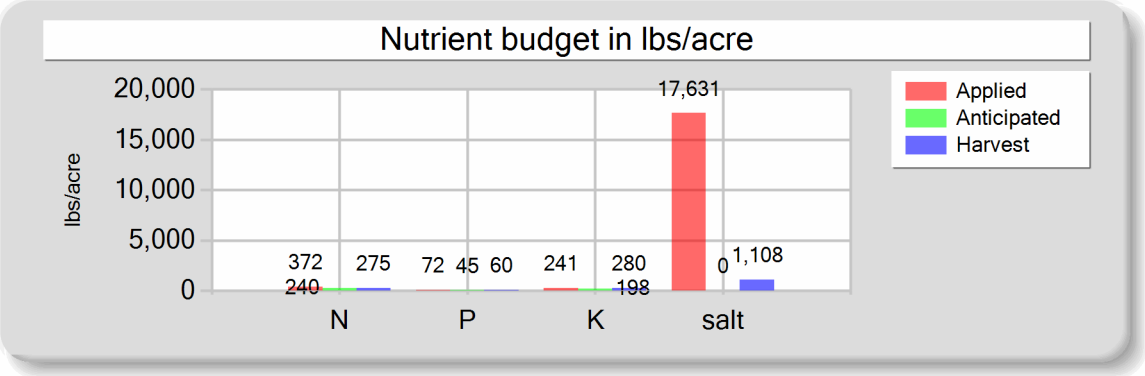
Fresh water applied	
12,825,000.00	*gallons*
472.30	*acre-inches*
11.81	*inches/acre*

Process wastewater applied	
334,578.84	*gallons*
12.32	*acre-inches*
0.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

#2 - 05/25/2023: Corn, silage

Field name: #2 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	351.35	71.61	240.72	378.50
Fresh water	13.41	0.00	0.00	17,252.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	371.77	71.61	240.72	17,631.05
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	275.27	60.42	280.30	1,107.79
Nutrient balance	96.50	11.19	-39.58	16,523.27
Applied to removed ratio	1.35	1.19	0.86	15.92

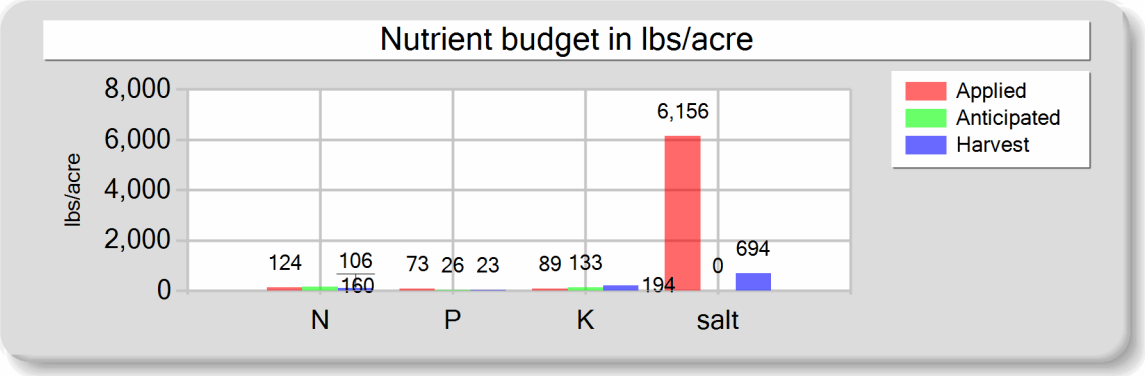
Fresh water applied	
42,300,000.00	*gallons*
1,557.77	*acre-inches*
38.94	*inches/acre*

Process wastewater applied	
725,698.54	*gallons*
26.73	*acre-inches*
0.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

#3 - 11/04/2022: Oats, silage-soft dough

Field name: #3 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	112.92	72.95	89.46	302.80
Fresh water	4.55	0.00	0.00	5,853.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	124.47	72.95	89.46	6,156.17
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	106.17	23.12	194.36	693.52
Nutrient balance	18.30	49.83	-104.89	5,462.65
Applied to removed ratio	1.17	3.16	0.46	8.88

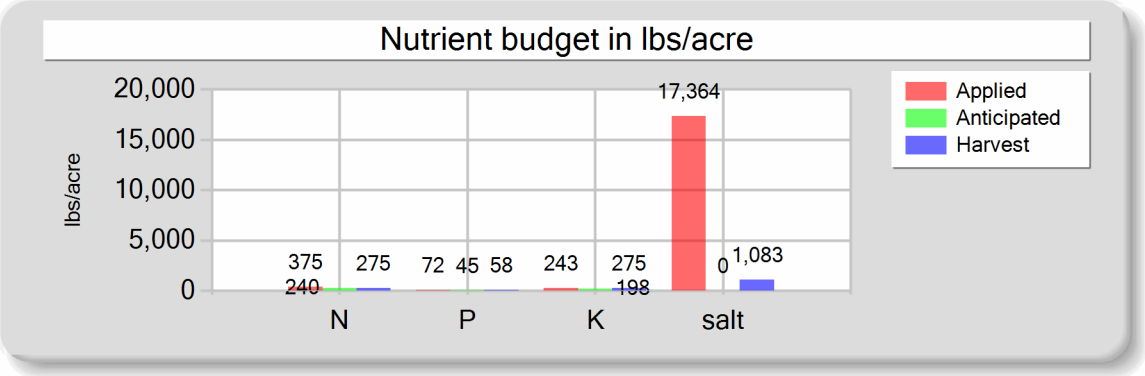
Fresh water applied	
13,275,000.00	*gallons*
488.87	*acre-inches*
13.21	*inches/acre*

Process wastewater applied	
305,128.97	*gallons*
11.24	*acre-inches*
0.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

#3 - 05/25/2023: Corn, silage

Field name: #3 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	355.16	72.39	243.34	382.60
Fresh water	13.20	0.00	0.00	16,981.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	375.37	72.39	243.34	17,363.99
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	274.96	58.33	274.96	1,083.18
Nutrient balance	100.40	14.06	-31.63	16,280.80
Applied to removed ratio	1.37	1.24	0.88	16.03

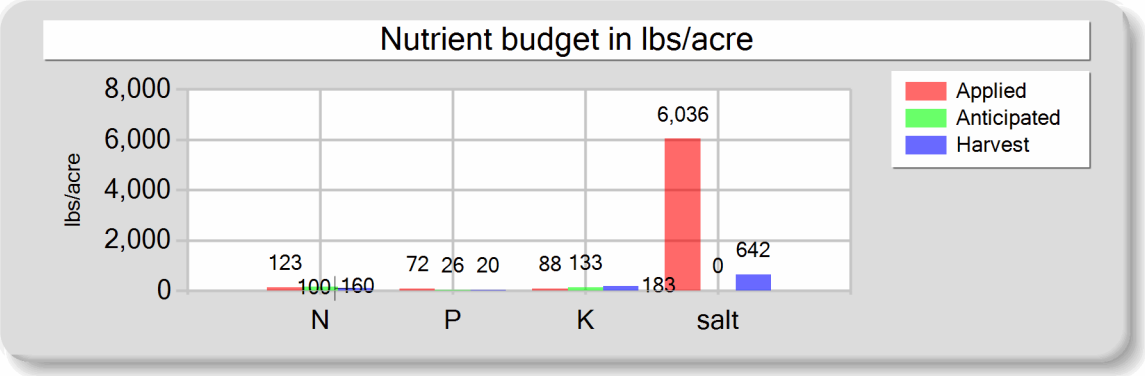
Fresh water applied	
38,512,500.00	*gallons*
1,418.28	*acre-inches*
38.33	*inches/acre*

Process wastewater applied	
678,552.33	*gallons*
24.99	*acre-inches*
0.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

#4 - 11/05/2022: Oats, silage-soft dough

Field name: #4 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	111.22	71.85	88.12	298.26
Fresh water	4.46	0.00	0.00	5,737.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	122.68	71.85	88.12	6,035.88
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	99.54	19.74	183.45	641.65
Nutrient balance	23.15	52.11	-95.32	5,394.24
Applied to removed ratio	1.23	3.64	0.48	9.41

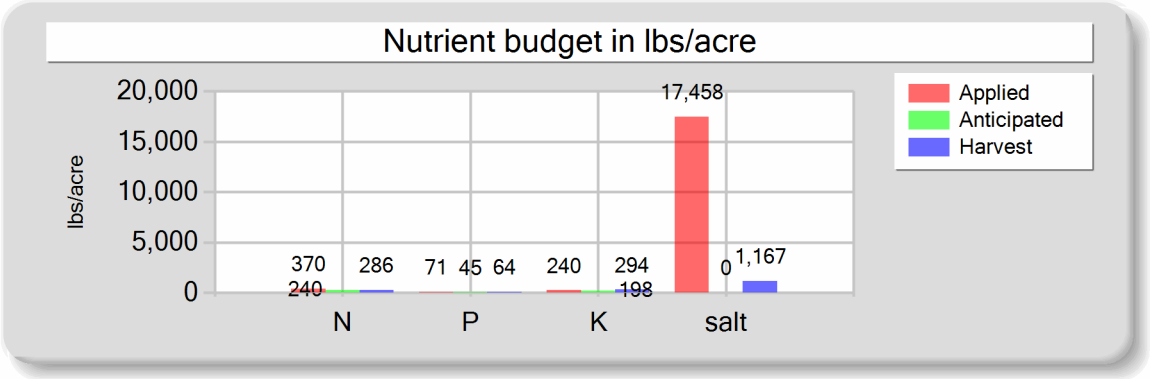
Fresh water applied	
13,012,500.00	*gallons*
479.21	*acre-inches*
12.95	*inches/acre*

Process wastewater applied	
300,548.49	*gallons*
11.07	*acre-inches*
0.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

#4 - 05/26/2023: Corn, silage

Field name: #4 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	349.89	71.31	239.72	376.92
Fresh water	13.28	0.00	0.00	17,080.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	370.17	71.31	239.72	17,457.52
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	285.62	64.44	294.33	1,166.87
Nutrient balance	84.55	6.88	-54.61	16,290.65
Applied to removed ratio	1.30	1.11	0.81	14.96

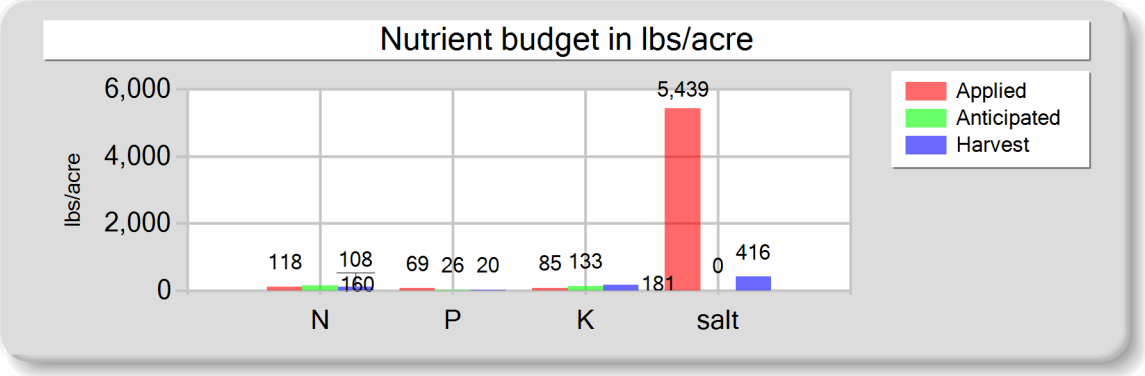
Fresh water applied	
38,737,500.00	*gallons*
1,426.57	*acre-inches*
38.56	*inches/acre*

Process wastewater applied	
668,481.66	*gallons*
24.62	*acre-inches*
0.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

#5 - 11/05/2022: Oats, silage-soft dough

Field name: #5 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	107.06	69.16	84.82	287.09
Fresh water	4.01	0.00	0.00	5,151.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	118.06	69.16	84.82	5,439.03
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	107.54	20.37	180.87	415.51
Nutrient balance	10.52	48.79	-96.05	5,023.52
Applied to removed ratio	1.10	3.40	0.47	13.09

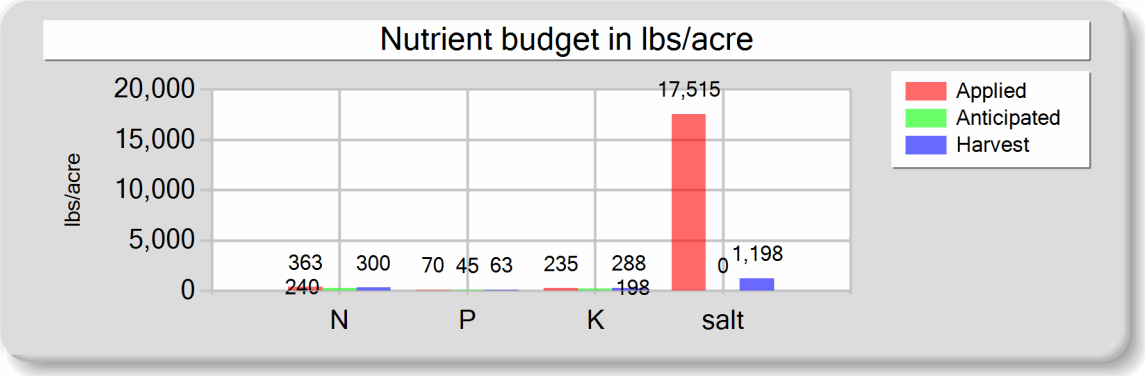
Fresh water applied	
12,000,000.00	*gallons*
441.92	*acre-inches*
11.63	*inches/acre*

Process wastewater applied	
297,114.75	*gallons*
10.94	*acre-inches*
0.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

#5 - 05/26/2023: Corn, silage

Field name: #5 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	342.55	69.82	234.70	369.02
Fresh water	13.33	0.00	0.00	17,146.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	362.88	69.82	234.70	17,515.32
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	300.47	62.53	288.31	1,198.40
Nutrient balance	62.41	7.29	-53.62	16,316.91
Applied to removed ratio	1.21	1.12	0.81	14.62

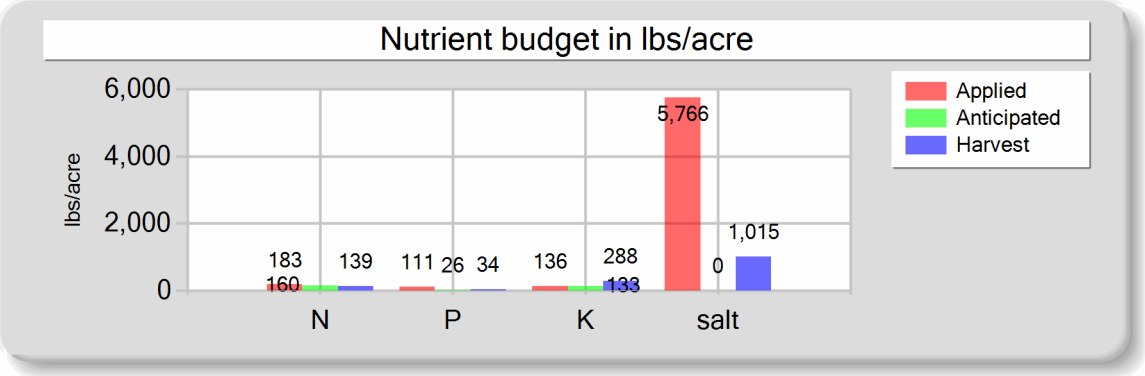
Fresh water applied	
39,937,500.00	*gallons*
1,470.76	*acre-inches*
38.70	*inches/acre*

Process wastewater applied	
672,148.30	*gallons*
24.75	*acre-inches*
0.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

#6 - 11/05/2022: Oats, silage-soft dough

Field name: #6 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	171.52	110.81	135.89	459.95
Fresh water	4.13	0.00	0.00	5,306.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	182.64	110.81	135.89	5,766.08
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	139.33	34.24	288.11	1,015.46
Nutrient balance	43.31	76.56	-152.21	4,750.61
Applied to removed ratio	1.31	3.24	0.47	5.68

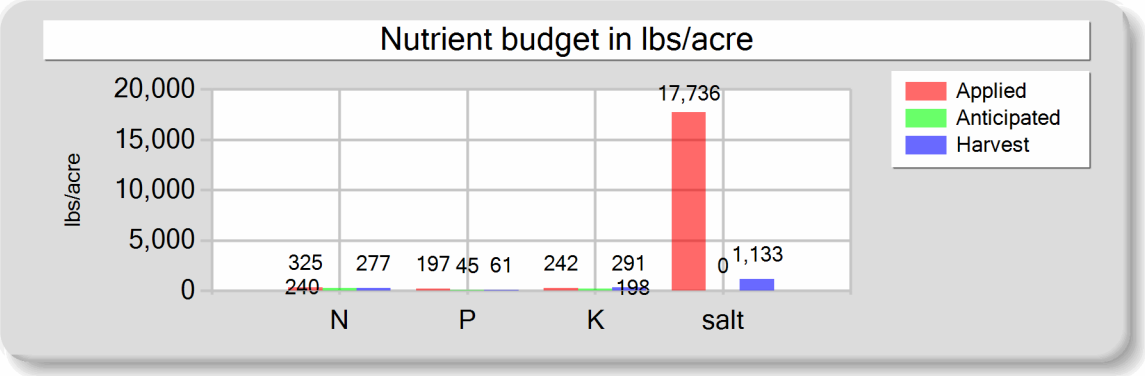
Fresh water applied	
16,912,500.00	*gallons*
622.83	*acre-inches*
11.98	*inches/acre*

Process wastewater applied	
651,380.28	*gallons*
23.99	*acre-inches*
0.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

#6 - 05/26/2023: Corn, silage

Field name: #6 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	304.92	196.98	241.58	817.67
Fresh water	13.15	0.00	0.00	16,918.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	325.07	196.98	241.58	17,736.09
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	277.31	60.87	290.84	1,132.93
Nutrient balance	47.76	136.11	-49.26	16,603.16
Applied to removed ratio	1.17	3.24	0.83	15.66

Fresh water applied	
53,925,000.00	*gallons*
1,985.87	*acre-inches*
38.19	*inches/acre*

Process wastewater applied	
1,157,983.88	*gallons*
42.64	*acre-inches*
0.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 93.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	3,300.00	21,300.00							0.00
DL	0.01	0.01	0.01							1.00

Solid Manure

Sample and source description: Solid Manure

Sample date: 11/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,600.00	3,100.00	16,200.00							0.00
DL	100.00	100.00	100.00							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,480.00	732.00		1.80	239.00	1,300.00								13,800.00	5,000
DL	1.00	0.50		0.10	0.10	0.50								10.00	10

Slurry

Sample and source description: Slurry

Sample date: 04/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2,320.00	751.00		0.70	473.00	1,590.00								14,400.00	2,500
DL	1.00	0.50		0.10	0.10	0.50								10.00	10

Slurry

Sample and source description: Slurry

Sample date: 09/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,640.00	521.00		0.80	1,060.00	1,300.00								10,100.00	4,400
DL	1.00	0.50		0.10	0.10	0.50								10.00	10

Q4 2023

Sample and source description: Q4 2023

Sample date: 11/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	102.00	38.30	0.00	0.70	33.40	331.00								1,870.00	1,550
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Dairy Well

Domestic Well - Dairy Well

Dairy Well

Sample description: Dairy Well

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			7.90								2,040.00	
DL			0.10								10.00	

Irrigation Water - Canal Water

Canal Water

Sample description: Canal Water

Sample date: 07/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00		0.00								32.00	33
DL	0.01		0.10								10.00	10

Irrigation Water - East Well

East Well

Sample description: East Well

Sample date: 08/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.04		3.00								2,170.00	1,400
DL	1.00		0.10								10.00	10

Irrigation Water - North West Well

Irrigation Water - North West Well

North West Well

Sample description: North West Well

Sample date: 08/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00		0.00								3,420.00	2,510
DL	0.10		0.10								10.00	10

Irrigation Water - South West Well

South West Well

Sample description: South West Well

Sample date: 08/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00		0.00								3,360.00	2,400
DL	1.00		0.10								10.00	10

D. SOIL ANALYSES

#1

HFR 1

Sample and source description: HFR 1

Sample date: 05/24/2022 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			43.00				
DL			2.00				

#2

HFR 2

Sample and source description: HFR 2

Sample date: 05/24/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			63.00				
DL			2.00				

#3

HFR 3

Sample and source description: HFR 3

Sample date: 05/24/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			33.00				
DL			2.00				

#4

HFR 4

Sample and source description: HFR 4

Sample date: 05/24/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			34.00				
DL			2.00				

#5

#5

HFR 5

Sample and source description: HFR 5

Sample date: 05/24/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			49.00				
DL			2.00				

#6

HFR 6

Sample and source description: HFR 6

Sample date: 05/24/2022 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			52.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

#1 - 11/04/2022: Oats, silage-soft dough

1A

Sample and source description: 1A

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	2,500.00	22,700.00		9.10
DL	100.00	100.00	100.00		0.01

#1 - 11/04/2022: Oats, silage-soft dough

1B

Sample and source description: 1B

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,400.00	2,700.00	24,500.00		8.70
DL	100.00	100.00	100.00		0.01

#1 - 05/25/2023: Corn, silage

1A

Sample and source description: 1A

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,600.00	3,400.00	16,600.00		6.50
DL	100.00	100.00	100.00		0.01

1B

Sample and source description: 1B

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	3,500.00	17,000.00		6.60
DL	100.00	100.00	100.00		0.01

#2 - 11/04/2022: Oats, silage-soft dough

#2 - 11/04/2022: Oats, silage-soft dough

2

Sample and source description: 2

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,900.00	2,500.00	21,800.00		8.20
DL	100.00	100.00	100.00		0.01

#2 - 05/25/2023: Corn, silage

2

Sample and source description: 2

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	3,600.00	16,700.00		6.60
DL	100.00	100.00	100.00		0.01

#3 - 11/04/2022: Oats, silage-soft dough

3

Sample and source description: 3

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	2,700.00	22,700.00		8.10
DL	100.00	100.00	100.00		0.01

#3 - 05/25/2023: Corn, silage

3

Sample and source description: 3

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	3,500.00	16,500.00		6.50
DL	100.00	100.00	100.00		0.01

#4 - 11/05/2022: Oats, silage-soft dough

4

Sample and source description: 4

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,400.00	22,300.00		7.80
DL	100.00	100.00	100.00		0.01

#4 - 05/26/2023: Corn, silage

4

Sample and source description: 4

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	3,700.00	16,900.00		6.70
DL	100.00	100.00	100.00		0.01

#5 - 11/05/2022: Oats, silage-soft dough

5

Sample and source description: 5

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	2,500.00	22,200.00		5.10
DL	100.00	100.00	100.00		0.01

#5 - 05/26/2023: Corn, silage

5

Sample and source description: 5

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,300.00	3,600.00	16,600.00		6.90
DL	100.00	100.00	100.00		0.01

#6 - 11/05/2022: Oats, silage-soft dough

6A

Sample and source description: 6A

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,900.00	24,400.00		8.60
DL	100.00	100.00	100.00		0.01

#6 - 11/05/2022: Oats, silage-soft dough

6B

Sample and source description: 6B

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,900.00	3,000.00	25,700.00		8.70
DL	100.00	100.00	100.00		0.01

#6 - 05/26/2023: Corn, silage

6A

Sample and source description: 6A

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	3,600.00	17,200.00		6.70
DL	100.00	100.00	100.00		0.01

6B

Sample and source description: 6B

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,600.00	16,800.00		6.70
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

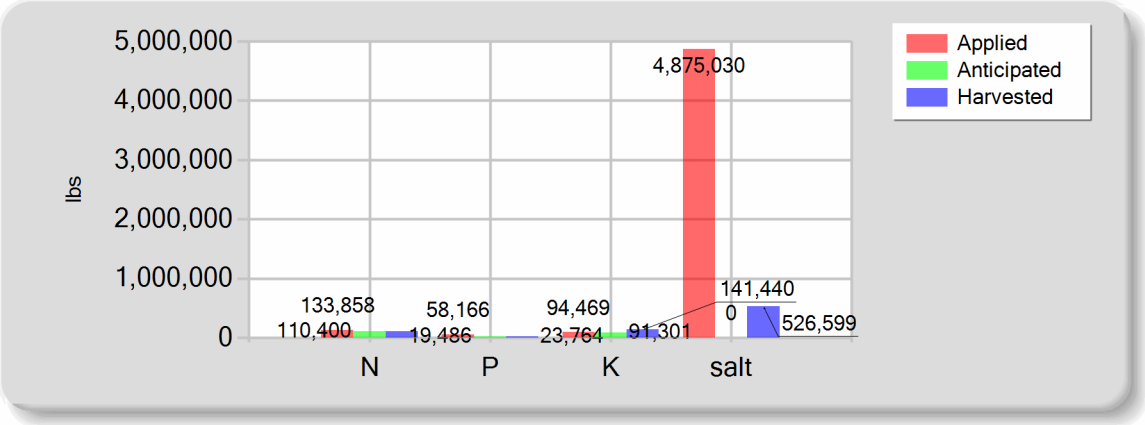
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

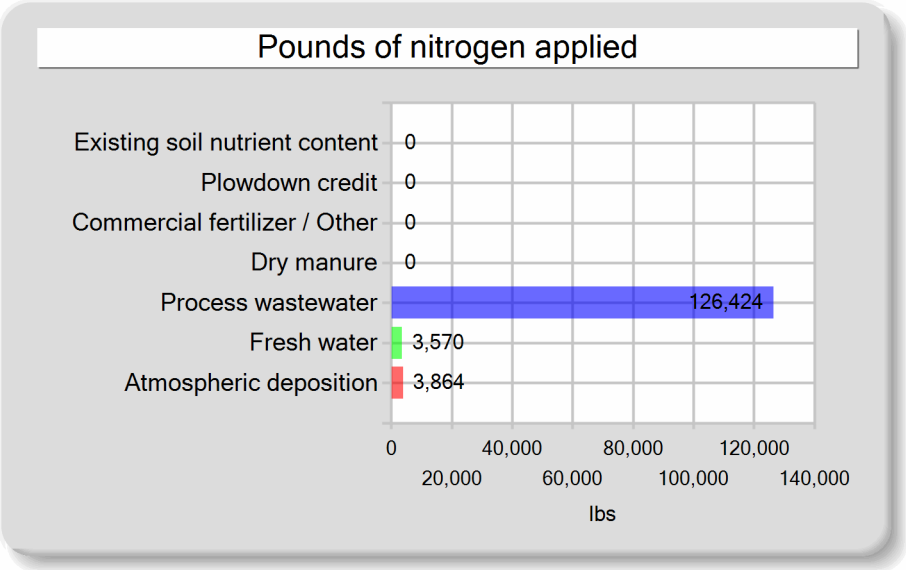
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

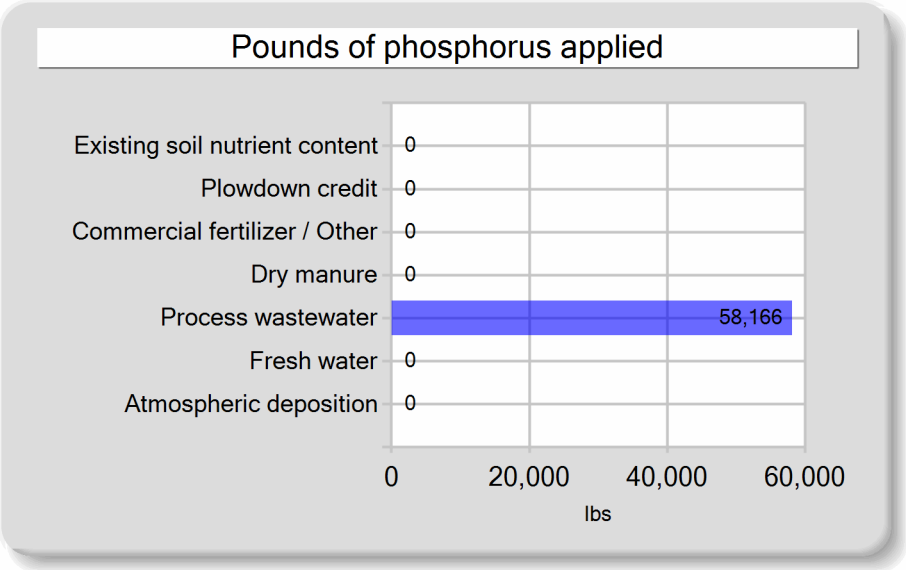
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	126,423.64	58,166.07	94,468.67	253,735.52
Fresh water	3,569.87	0.00	0.00	4,621,294.36
Atmospheric deposition	3,864.00	0.00	0.00	0.00
Total nutrients applied	133,857.50	58,166.07	94,468.67	4,875,029.87
Anticipated crop nutrient removal	110,400.00	19,485.60	91,300.80	0.00
Actual crop nutrient removal	107,735.78	23,764.34	141,440.24	526,599.08
Nutrient balance	26,121.72	34,401.73	-46,971.57	4,348,430.80
Applied to removed ratio	1.24	2.45	0.67	9.26

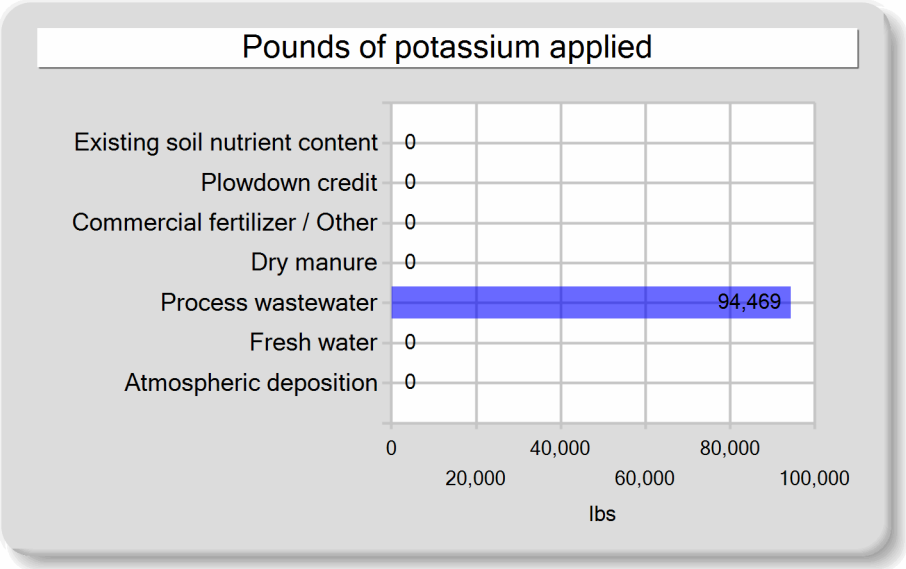
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

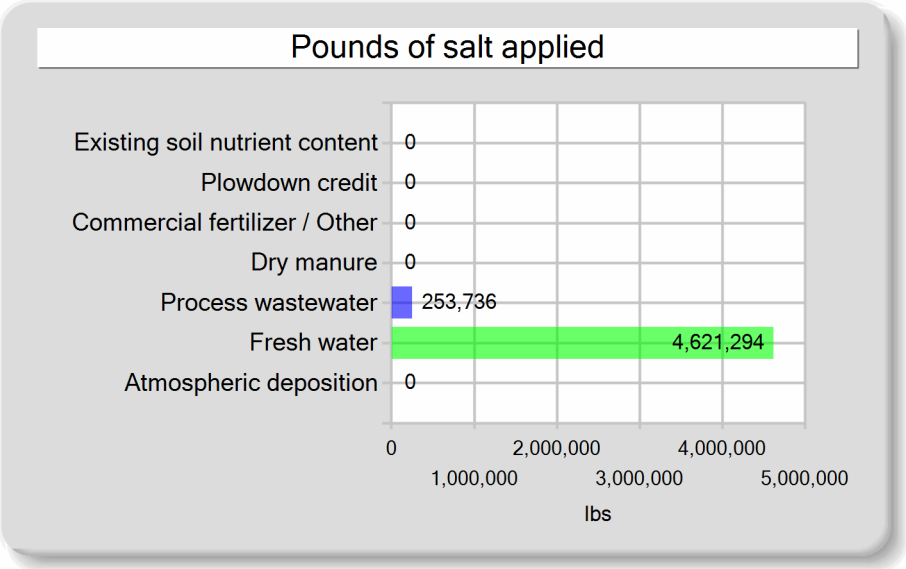


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

****Soluble phosphorus reported as OLSEN

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Bobby Borba
Bobby Borba (Jun 19, 2024 15:39 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Bob Borba	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/19/2024	
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dairy Central Heifers

Received: 08/15/2023 14:18
Reported: 08/22/2023 09:42

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H1368-01	East Well	Ag Water	T. Madruga		08/14/2023 14:00
23H1368-02	North West Well	Ag Water	T. Madruga		08/14/2023 15:00
23H1368-03	South West Well	Ag Water	T. Madruga		08/14/2023 16:00

Default Cooler Temperature on Receipt °C: 2.5
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dairy Central Heifers

Received: 08/15/2023 14:18
Reported: 08/22/2023 09:42

Sample Results

Sample: East Well
23H1368-01 (Water)

Sampled: 8/14/2023 14:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.17**	mmhos/cm	0.01	1		08/15/23 18:10	SM 2510 B		BEH0757
Electrical Conductivity umhos	**2170**	umhos/cm	10.0	1		08/15/23 18:10	SM 2510 B		BEH0757
Nitrate Nitrogen as NO3N	**3.0**	mg/L	0.1	1	10	08/16/23 02:03	EPA 300.0		BEH0724
pH	**7.7**	units	1.0	1		08/15/23 18:10	SM 4500-H+	H	BEH0757
Total Filterable Solids (TDS)	**1400**	mg/L	10.0	1		08/16/23 15:09	SM 2540 C		BEH0718
Temperature	**25.0**	°C	0.0	1		08/15/23 18:10	SM 2510 B		BEH0757
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/18/23 13:17	SM 4500-NH3 C		BEH0808
Total Nitrogen	**3.04**	mg/L	1.00	1		08/18/23 13:17	SM 4500-NH3 C		BEH0808

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dairy Central Heifers

Received: 08/15/2023 14:18
Reported: 08/22/2023 09:42

Sample Results
(Continued)

Sample: North West Well
23H1368-02 (Water)

Sampled: 8/14/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**3.42**	mmhos/cm	0.01	1		08/15/23 18:11	SM 2510 B		BEH0757
Electrical Conductivity umhos	**3420**	umhos/cm	10.0	1		08/15/23 18:11	SM 2510 B		BEH0757
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/16/23 02:23	EPA 300.0		BEH0724
pH	**7.7**	units	1.0	1		08/15/23 18:11	SM 4500-H+	H	BEH0757
Total Filterable Solids (TDS)	**2510**	mg/L	10.0	1		08/16/23 15:09	SM 2540 C		BEH0718
Temperature	**25.0**	°C	0.0	1		08/15/23 18:11	SM 2510 B		BEH0757
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/18/23 13:19	SM 4500-NH3 C		BEH0808
Total Nitrogen	ND	mg/L	1.00	1		08/18/23 13:19	SM 4500-NH3 C		BEH0808

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dairy Central Heifers

Received: 08/15/2023 14:18
Reported: 08/22/2023 09:42

Sample Results
(Continued)

Sample: South West Well
23H1368-03 (Water)

Sampled: 8/14/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**3.36**	mmhos/cm	0.01	1		08/15/23 18:12	SM 2510 B		BEH0757
Electrical Conductivity umhos	**3360**	umhos/cm	10.0	1		08/15/23 18:12	SM 2510 B		BEH0757
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/16/23 02:43	EPA 300.0		BEH0724
pH	**7.6**	units	1.0	1		08/15/23 18:12	SM 4500-H+	H	BEH0757
Total Filterable Solids (TDS)	**2400**	mg/L	10.0	1		08/16/23 15:09	SM 2540 C		BEH0718
Temperature	**25.0**	°C	0.0	1		08/15/23 18:12	SM 2510 B		BEH0757
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/18/23 13:20	SM 4500-NH3 C		BEH0808
Total Nitrogen	ND	mg/L	1.00	1		08/18/23 13:20	SM 4500-NH3 C		BEH0808

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Ranch: Dairy Central Heifers

Received: 08/15/2023 14:18
Reported: 08/22/2023 09:42

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0718										
Blank (BEH0718-BLK1)					Prepared: 8/15/2023	Analyzed: 8/16/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEH0718-BS1)					Prepared: 8/15/2023	Analyzed: 8/16/2023				
Total Filterable Solids (TDS)	22.5		10.0	mg/L	2000		1.12	0-200		
Duplicate (BEH0718-DUP1)		**Source: 23H0090-01**			Prepared: 8/15/2023	Analyzed: 8/16/2023				
Total Filterable Solids (TDS)	2540		10.0	mg/L		2420			4.84	10
Duplicate (BEH0718-DUP2)		**Source: 23H1316-03**			Prepared: 8/15/2023	Analyzed: 8/16/2023				
Total Filterable Solids (TDS)	310		10.0	mg/L		290			6.67	10
Reference (BEH0718-SRM1)					Prepared: 8/15/2023	Analyzed: 8/16/2023				
Total Filterable Solids (TDS)	320			mg/L	325.0		98.5	90-110		
Reference (BEH0718-SRM2)					Prepared: 8/15/2023	Analyzed: 8/16/2023				
Total Filterable Solids (TDS)	487			mg/L	495.0		98.3	90-110		

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dairy Central Heifers

Received: 08/15/2023 14:18
Reported: 08/22/2023 09:42

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0724										
Blank (BEH0724-BLK1)					Prepared & Analyzed: 8/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0724-BLK2)					Prepared & Analyzed: 8/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0724-BLK3)					Prepared: 8/15/2023 Analyzed: 8/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0724-BS1)					Prepared & Analyzed: 8/15/2023					
Nitrate Nitrogen as NO3N	5.5		0.1	mg/L	5.000		110	90-110		
LCS (BEH0724-BS2)					Prepared: 8/15/2023 Analyzed: 8/16/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
Duplicate (BEH0724-DUP1)		**Source: 23H1360-01**			Prepared & Analyzed: 8/15/2023					
Nitrate Nitrogen as NO3N	0.1		0.1	mg/L		0.08			26.2	10
Duplicate (BEH0724-DUP2)		**Source: 23H1386-04**			Prepared: 8/15/2023 Analyzed: 8/16/2023					
Nitrate Nitrogen as NO3N	0.7		0.1	mg/L		0.7			0.542	10
Matrix Spike (BEH0724-MS1)		**Source: 23H1360-01**			Prepared & Analyzed: 8/15/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000	0.08	97.4	90-110		
Matrix Spike (BEH0724-MS2)		**Source: 23H1386-04**			Prepared: 8/15/2023 Analyzed: 8/16/2023					
Nitrate Nitrogen as NO3N	5.7		0.1	mg/L	5.000	0.7	99.3	90-110		
Reference (BEH0724-SRM1)					Prepared & Analyzed: 8/15/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		98.0	90-110		
Reference (BEH0724-SRM2)					Prepared: 8/15/2023 Analyzed: 8/16/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		98.6	90-110		
Reference (BEH0724-SRM3)					Prepared: 8/15/2023 Analyzed: 8/16/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.2	90-110		

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Ranch: Dairy Central Heifers

Received: 08/15/2023 14:18
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0757										
Blank (BEH0757-BLK1)					Prepared & Analyzed: 8/15/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.3		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0757-BLK2)					Prepared & Analyzed: 8/15/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.1		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0757-BLK3)					Prepared & Analyzed: 8/15/2023					
pH	7.0		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH0757-DUP1)		**Source: 23H1364-01**			Prepared & Analyzed: 8/15/2023					
Electrical Conductivity	0.24		0.01	mmhos/cm		0.24			1.49	10
pH	7.8		1.0	units		7.8			0.384	10
Electrical Conductivity umhos	240		10.0	umhos/cm		244			1.49	10
Duplicate (BEH0757-DUP2)		**Source: 23H1368-03**			Prepared & Analyzed: 8/15/2023					
Electrical Conductivity	3.35		0.01	mmhos/cm		3.36			0.215	10
pH	7.6		1.0	units		7.6			0.656	10
Electrical Conductivity umhos	3350		10.0	umhos/cm		3360			0.215	10
Reference (BEH0757-SRM1)					Prepared & Analyzed: 8/15/2023					
Electrical Conductivity	515			umhos/cm	538.0		95.7	90-110		
Reference (BEH0757-SRM2)					Prepared & Analyzed: 8/15/2023					
pH	5.8			units	5.820		100	28178-101.71		
Reference (BEH0757-SRM3)					Prepared & Analyzed: 8/15/2023					
Electrical Conductivity	949			umhos/cm	1000		94.9	90-110		
Electrical Conductivity umhos	949			umhos/cm	1000		94.9	90-110		
Reference (BEH0757-SRM4)					Prepared & Analyzed: 8/15/2023					
Electrical Conductivity	954			umhos/cm	1000		95.4	90-110		
Electrical Conductivity umhos	954			umhos/cm	1000		95.4	90-110		
Reference (BEH0757-SRM5)					Prepared & Analyzed: 8/15/2023					
Electrical Conductivity	958			umhos/cm	1000		95.8	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0757 (Continued)										
Reference (BEH0757-SRM5)					Prepared & Analyzed: 8/15/2023					
Electrical Conductivity umhos	958			umhos/cm	1000		95.8	90-110		
Reference (BEH0757-SRM6)					Prepared & Analyzed: 8/15/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0757-SRM7)					Prepared & Analyzed: 8/15/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0757-SRM8)					Prepared & Analyzed: 8/15/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0808										
Blank (BEH0808-BLK1)					Prepared: 8/16/2023	Analyzed: 8/18/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEH0808-BLK2)					Prepared: 8/16/2023	Analyzed: 8/18/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEH0808-BS1)					Prepared: 8/16/2023	Analyzed: 8/18/2023				
Kjeldahl Nitrogen (TKN), Total	5.65		1.00	mg/L	5.709		99.0	90-110		
LCS (BEH0808-BS2)					Prepared: 8/16/2023	Analyzed: 8/21/2023				
Kjeldahl Nitrogen (TKN), Total	6.17		1.00	mg/L	5.709		108	90-110		
Duplicate (BEH0808-DUP1)		**Source: 23H1173-02**			Prepared: 8/16/2023	Analyzed: 8/18/2023				
Kjeldahl Nitrogen (TKN), Total	16.8		1.40	mg/L		ND				10
Duplicate (BEH0808-DUP2)		**Source: 23H1455-02**			Prepared: 8/16/2023	Analyzed: 8/21/2023				
Kjeldahl Nitrogen (TKN), Total	284		3.50	mg/L		287			1.23	10
Matrix Spike (BEH0808-MS1)		**Source: 23H1173-02**			Prepared: 8/16/2023	Analyzed: 8/18/2023				
Kjeldahl Nitrogen (TKN), Total	8.86		1.40	mg/L	7.992	ND	111	90-110		
Matrix Spike (BEH0808-MS2)		**Source: 23H1455-02**			Prepared: 8/16/2023	Analyzed: 8/21/2023				
Kjeldahl Nitrogen (TKN), Total	278		3.50	mg/L	9.990	287	NR	90-110		
Reference (BEH0808-SRM1)					Prepared: 8/16/2023	Analyzed: 8/21/2023				
Kjeldahl Nitrogen (TKN), Total	24.3			mg/L	23.80		102	90-110		

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08/15/23 14:18 23H1368

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 3 No. of Bottles: 6

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Dairy Central Heifers
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

( ) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
(X) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	East Well	8/14/2023	1400	2.5	
2	North West Well	8/14/2023	1500	2.8	
3	South West Well	8/14/2023	1600	1.5	
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	8/14/2023 @ 1600	8/15/2023 @ 900
Second	JLF	Dellavalle Laboratory, Inc.	8/15/2023 @ 900	8-15-23 1:55
Third				
Fourth	JLF	DLF	8/15/23 14:18	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

08/15/23 14:18 23H1368

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)	1	1	1							
1 L unpreserved (White)	1	1	1							
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dairy Central Heifers

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:27

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0675-01	Dairy Well	Ag Water	T. Madruga		11/13/2023 16:00

Default Cooler Temperature on Receipt °C: 3.8
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Sample Results

Sample: Dairy Well
23K0675-01 (Water)

Sampled: 11/13/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.04**	mmhos/cm	0.01	1		11/16/23 13:04	SM 2510 B		BEK0477
Electrical Conductivity umhos	**2040**	umhos/cm	10.0	1		11/16/23 13:04	SM 2510 B		BEK0477
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 16:00	Field		BEK0420
Nitrate Nitrogen as NO3N	**7.9**	mg/L	0.1	1	10	11/15/23 00:39	EPA 300.0		BEK0412
Temperature	**25.0**	units	0.0	1		11/16/23 13:04	SM 4500-H+	H	BEK0477
pH	**7.8**	units	1.0	1		11/16/23 13:04	SM 4500-H+	H	BEK0477

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0412										
Blank (BEK0412-BLK1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0412-BLK2)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0412-BLK3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0412-BS1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		100	90-110		
LCS (BEK0412-BS2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		100	90-110		
Duplicate (BEK0412-DUP1)		**Source: 23K0676-01**			Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	15.7		0.1	mg/L		15.5			0.802	10
Duplicate (BEK0412-DUP2)		**Source: 23K0676-02**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	36.4		0.1	mg/L		36.4			0.0192	10
Matrix Spike (BEK0412-MS1)		**Source: 23K0676-01**			Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	20.6		0.1	mg/L	5.000	15.5	102	90-110		
Matrix Spike (BEK0412-MS2)		**Source: 23K0676-02**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	42.1		0.1	mg/L	5.000	36.4	113	90-110		
Reference (BEK0412-SRM1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.9	90-110		
Reference (BEK0412-SRM2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		98.7	90-110		
Reference (BEK0412-SRM3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.2	90-110		

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Received: 11/14/2023 15:45
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0477										
Blank (BEK0477-BLK1)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEK0477-BLK2)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.5		1.0	units						
Blank (BEK0477-BLK3)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.3		1.0	units						
Duplicate (BEK0477-DUP1)		**Source: 23K0667-01**			Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	4.04		0.01	mmhos/cm		4.04			0.0495	10
pH	7.8		1.0	units		7.6			1.82	10
Electrical Conductivity umhos	4040		10.0	umhos/cm		4040			0.0495	10
Duplicate (BEK0477-DUP2)		**Source: 23K0676-03**			Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	1.35		0.01	mmhos/cm		1.36			0.791	10
Electrical Conductivity umhos	1350		10.0	umhos/cm		1360			0.791	10
pH	7.6		1.0	units		7.6			0.264	10
Reference (BEK0477-SRM1)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	578			umhos/cm	538.0		107	90-110		
Reference (BEK0477-SRM2)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0477-SRM3)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0477-SRM4)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0477-SRM5)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dairy Central Heifers

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:27

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0477 (Continued)										
Reference (BEK0477-SRM5)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0477-SRM6)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0477-SRM7)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0477-SRM8)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/14/23 15:45 23K0675

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

JG

No. Samples 1 No. of Bottles: 1

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com

COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Dairy Central Heifers
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents

Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Dairy Well	11/13/2023	1600	3.8	ND
2					
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/13/2023 @ 1600	11/14/2023 @ 900
Second	JCP	Dellavalle Laboratory, Inc.	11/14/2023 @ 900	11-14-23 3:30
Third				
Fourth		DLL	11/14 15:45	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/14/23 15:45 23K0675

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1									
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										